

HOW A CAMEL VIRUS BECAME A GLOBAL ONE HEALTH THREAT

A One Health GIS Analysis of MERS-CoV Risk Pathways

1 INTRODUCTION

Good day everyone,

Today, my team and I want to take you on a journey across continents through animal reservoirs, livestock trade routes, mass gatherings, and global aviation networks to answer one important question:

How did a virus associated with camels become a global public health threat?

This is the story of the Middle East Respiratory Syndrome coronavirus, also known as - **MERS-CoV**.

MERS is a zoonotic disease, meaning it spreads from animals to humans. It was first identified in Saudi Arabia in 2012 and is associated with severe respiratory illness and significant mortality.

What makes MERS especially important is not only how dangerous it can be, but how it sits at the intersection of animal health, human behaviour, global mobility, and environmental systems.

This makes MERS a classic One Health challenge.

For this project, we applied Geographic Information Systems and Python-based spatial analysis to investigate how camel reservoirs, livestock trade networks, human mobility, and air travel interact to shape MERS transmission pathways.

We used three major datasets:

1. WHO global MERS case data
2. FAOSTAT camel trade data
3. ICAO international aviation data

Let us now follow the transmission pathway.

First, we begin with the animal reservoir.

Dromedary camels are the primary reservoir of MERS-CoV.

Our analysis identified major camel reservoir hotspots in the Horn of Africa and the Arabian Peninsula.

This means the virus existed in animal populations long before widespread human recognition.

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The next question becomes:

How does risk move geographically?

This brings us to camel trade.

Our camel trade analysis revealed that the major exporters were:

Somalia, Sudan, Djibouti

Most notably, Somalia emerged as the largest camel exporter.

This suggests that camel trade networks may play a critical role in moving infected animal populations across borders.

In simple terms, risk moves with trade.

Next, we arrive at the most important human transmission hotspot.

Our human MERS case analysis identified Saudi Arabia as the principal hotspot of reported human infections.

This was one of the strongest findings in the project.

Why is this important?

Because this region represents a major convergence point of:

- Camel exposure
- Active trade
- Human-animal interaction
- Mass gathering events

This creates ideal conditions for spillover and transmission.

Now let us consider mass gatherings.

Each year, millions of people travel to Saudi Arabia for Hajj and Umrah.

These represent some of the largest recurring mass gatherings in the world.

From a public health perspective, this creates an important question:

What happens when potentially exposed individuals return home?

This brings us to international air mobility.

Our aviation network analysis identified the strongest mobility corridor as:

India → United Arab Emirates

with approximately **8,719,069 passengers.**

This was one of the most striking discoveries in the project.

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We also found strong aviation connectivity with:

United Arab Emirates, Saudi Arabia, Qatar, Kuwait

This means millions of passengers move annually through routes connected to the Arabian Peninsula.

And this is where local outbreaks can become global threats.

A virus originating in camels can move through trade systems, amplify in human populations, and spread internationally through aviation networks.

One of the most important real-world examples occurred in South Korea in 2015.

A single traveler returning from the Middle East introduced MERS into South Korea.

What followed became the largest MERS outbreak recorded outside the Arabian Peninsula.

The outbreak resulted in 186 confirmed cases and 38 deaths.

This event demonstrated how rapidly a localized zoonotic threat can become an international public health emergency.

If Saudi Arabia represents the epicentre of human amplification, South Korea represents the clearest evidence of global dissemination.

This is the core message of our One Health framework.

- Camel reservoirs.
- Livestock trade.
- Human transmission hotspots.
- Mass gatherings.
- Global air mobility.
- International dissemination.

This pathway explains how a local zoonotic threat can become an international public health challenge.

Our findings highlight several important lessons.

- First, MERS cannot be understood by looking at human infections alone.
- Second, animal reservoirs and livestock trade are critical components of disease spread.
- Third, international aviation creates opportunities for long-distance dissemination.
- Finally, MERS demonstrates why One Health surveillance is essential.

Our findings also point toward important public health recommendations.

- First, Strengthen surveillance at the animal-human interface.
- Second, Improve monitoring of camel trade and livestock mobility.
- Third, Enhance surveillance during Hajj and Umrah mass gatherings.
- Finally, Strengthen international travel screening and rapid response systems.

Together, these strategies can significantly improve preparedness against future zoonotic outbreaks.

Our main conclusion is simple:

MERS spread is driven by interconnected animal, trade, and mobility systems.

The broader lesson extends beyond MERS.

In our increasingly connected world, the next outbreak may begin locally but spread globally.

Understanding these transmission pathways before crises occur is one of the strongest tools we have in public health preparedness.

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Check our reference table for more clarifications.

Thank you.